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/-

Giza.lean — the A⁵O Clock Standard against the Great Pyramid (exploratory).

Everything here is arithmetic on stated constants. The pyramid figures are the standard survey values (Petrie 1883; Lehner) rounded to whole units:

royal cubit = 7 palms = 28 fingers (≈ 523.6 mm)

Great Pyramid base 440 cubits, height 280 cubits, perimeter 1760

"Giza pi" perimeter / height = $1760 / 280 = 44/7 = 2 \cdot 22/7$

Clock constants are the sealed ones: period 17 500 ps, frequency 400/7 MHz.

Core Lean 4 only; every theorem zero axioms. Reading of the results is overlay and is labelled as such in the comments.

-/

namespace A5O.Giza

/-! ## Constants -/

```
def periodPs    : Nat := 17500      -- one tick, picoseconds
def freqNum     : Nat := 400 * 10 ^ 6 -- 400/7 MHz as a fraction
def freqDen     : Nat := 7
def secondPs    : Nat := 10 ^ 12
```

```
def palmsPerCubit : Nat := 7
def fingersPerCubit : Nat := 28
def baseCubits    : Nat := 440
def heightCubits   : Nat := 280
def perimeterCubits : Nat := 4 * baseCubits
```

/-! ## 1. Seven is the cubit's divisor and the clock's denominator. -/

```
theorem cubit_denominator : palmsPerCubit = freqDen := rfl
```

```
theorem fingers_are_four_palms : fingersPerCubit = 4 * palmsPerCubit := rfl
```

/-! ## 2. The pyramid's height and base are multiples of the same 40. -/

```
theorem height_is_forty_sevens : heightCubits = 40 * 7 := rfl
theorem base_is_forty_elevens  : baseCubits   = 40 * 11 := rfl
```

/-- Overlay (a): the clock numerator is that same 40, times ten. -/

```
theorem clock_numerator_is_forty_tens : 400 = 40 * 10 := rfl
```

/-! ## 3. Giza pi. perimeter / height = $44/7 = 2 \cdot (22/7)$. -/

theorem giza_pi : perimeterCubits * 7 = 44 * heightCubits := by decide

/-- 22/7 and 400/7 have the SAME fractional part: $400 = 7 \cdot 54 + 22$, so
 $400/7 = 54 + 22/7$. The clock frequency in MHz is Giza-pi plus 54.

(Both numerators are $\equiv 1 \pmod{7}$, hence both tails are 0.142857...) -/

theorem clock_is_giza_pi_plus_54 : $400 = 7 * 54 + 22$:= by decide

theorem both_residue_one : $400 \% 7 = 1 \wedge 22 \% 7 = 1$:= by decide

/-! ## 4. The period is seven remainders. $17\,500 = 7 \cdot 2\,500$. -/

theorem period_is_seven_remainders : periodPs = 7 * (secondPs % periodPs) := by decide

/-- Consequently the remainder needs exactly seven seconds to add up to one
 tick — the "seventh second resolves" theorem in ClockStandard.lean,
 seen from the other side. -/

theorem seven_remainders_make_a_tick : 7 * (secondPs % periodPs) = periodPs := by decide

/-! ## 5. Light in one tick $\approx$ ten royal cubits (the one physical relation).

$c = 299\,792\,458$ m/s (exact by definition of the metre).

$c \cdot T = 299\,792\,458 \cdot 17.5$ ns = 5.246 368... m.

Ten royal cubits at 523.6 mm = 5.236 m.

Stated as integer bounds in micrometres so the kernel can decide it:

the deviation is under 0.2 %. -/

def cPerSec_um : Nat := 299792458 * 10 ^ 6 -- micrometres per second

def lightPerTick_um : Nat := cPerSec_um * periodPs / secondPs -- = 5 246 368 μ m (floor)

def cubit_um : Nat := 523600 -- Petrie 20.62 in $\approx$ 523.7 mm; 523.6 used

theorem light_per_tick_value : lightPerTick_um = 5246368 := by decide

/-- 10 cubits < light-per-tick < 10.02 cubits. -/

theorem light_per_tick_is_ten_cubits :

$10 * cubit_um < lightPerTick_um \wedge 1000 * lightPerTick_um < 10020 * cubit_um$:= by
 decide

/-- Overlay (b): with light-per-tick taken as ten cubits, the pyramid's
 dimensions become tick counts: height 28 ticks, base 44 ticks,
 perimeter 176 ticks — and $176/28 = 44/7$ is Giza pi again. -/

theorem height_in_ticks : heightCubits = 28 * 10 := rfl

theorem base_in_ticks : baseCubits = 44 * 10 := rfl

theorem perimeter_in_ticks : perimeterCubits = 176 * 10 := rfl

theorem tick_ratio_is_giza_pi : $176 * 7 = 44 * 28$:= by decide

```
-- Overlay (c): 28 — the height in ticks — is the number of fingers in a cubit. -/  
theorem height_ticks_eq_fingers : heightCubits / 10 = fingersPerCubit := by decide
```

```
end A5O.Giza
```

```
#print axioms A5O.Giza.cubit_denominator  
#print axioms A5O.Giza.giza_pi  
#print axioms A5O.Giza.clock_is_giza_pi_plus_54  
#print axioms A5O.Giza.period_is_seven_remainders  
#print axioms A5O.Giza.light_per_tick_value  
#print axioms A5O.Giza.light_per_tick_is_ten_cubits  
#print axioms A5O.Giza.tick_ratio_is_giza_pi  
#print axioms A5O.Giza.height_ticks_eq_fingers
```